

Note: I found these screen shots in a facebook group.

Differential Equations (MTH401)

Question: 1 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

An infinite series of $(x-a)$ in the form of $c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$ where the coefficients $c_0, c_1, c_2, \dots$ and a are constants and x is a variable is called

Choices:

Real series

☐

Analytic series

☐

Power series

☐

None of them

☐

Differential Equations (MTH401)

Question: 2 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

The differential equation $(x^2 - 1)y'' + 2xy' + 6y = 0$ has singular point at

Choices:

 $x = 0$ ☐ $x = 1$ ☐ $x = \pm 1$ ☐ $x = -1$ ☐

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Differential Equations (MTH401)

Question: 3 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

If L denote the linear differential operators with constant coefficients, then $L_1 L_4 - L_2 L_3$ represents the

Choices:

☐ $\begin{vmatrix} L_1 & L_2 \\ L_4 & L_3 \end{vmatrix}$

☐ $\begin{vmatrix} L_1 & L_3 \\ L_4 & L_2 \end{vmatrix}$

☒ $\begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix}$

None of them

Differential Equations (MTH401)

Question: 4 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

The solution of the linear first order differential equation $\frac{dy}{dx} - 2y = 0$ is

Choices:

☒ (a) $y = e^{2x+c}$

☐ (b) $y = \sum_{n=0}^{\infty} \frac{x^n}{4n!}$

Both (a) and (b)

None of them

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Differential Equations (MTH401)

Question: **5** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

If wroskian of the solution vectors X_1 & X_2 is zero, then vectors are

Choices:

Linearly Independent

☐

Linearly dependent

☒

None of them

☐

Parallel

☐

Differential Equations (MTH401)

Question: **6** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

A singular point $x = x_0$ of the given equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ is said to be a regular singular point if

Choices:

$(x - x_0)P(x)$ is analytic at x_0

☐

$(x - x_0)Q(x)$ is analytic at x_0

☐

$(x - x_0)P(x)$ & $(x - x_0)^2Q(x)$ are analytic at x_0

☒

None of them

☐

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Differential Equations (MTH401)

Question: **7** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

$$\frac{dy}{dt} = 2x, \frac{dx}{dt} = 3y$$

.....& are dependent variables in differential equations

Choices:

x, t

☐

y, t

☐

x, y

☐

t

☐

Differential Equations (MTH401)

Question: **8** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

$$x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 10y = 0$$

For auxiliary equation of the D.E ,substitute

Choices:

$y = x^m$

☐

$y = -x$

☐

$y = x$

☐

$y = x^2$

☐

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Differential Equations (MTH401)

Question: 9 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

If $y = e^{-6x}$, then which of the following is the most accurate option?

Choices:

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)^2$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)^3$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D - 6)$.

☐ The annihilator operator of $y = e^{-6x}$ is $(D + 6)$.

Differential Equations (MTH401)

Question: 10 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

The Differential Equation $(x^2 - 4)y'' - 10xy' + y = 0$ has singularity at

Choices:

☐ $x = \pm 1$

☐ $x = \pm 2$

☐ $x = \pm 3$

☐ $x = \pm 4$

(a) The singular points of the equation $(x^2 - 1)y'' + 2xy' + 6y = 0$ are the solutions of $x^2 - 1 = 0$ or $x = \pm 1$. All other finite values of x are the ordinary points.

(b) The singular points need not be real numbers.

The equation $(x^2 + 1)y'' + 2xy' + 6y = 0$ has the singular points at the solutions of $x^2 + 1 = 0$, namely, $x = \pm i$.

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Differential Equations (MTH401)

Question: **11** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - \frac{1}{25}) = 0$$

The general solution of the equation is

Choices:

☐ $y = c_1 J_{\frac{1}{3}}(x) + c_2 J_{-\frac{1}{3}}(x)$

☐ $y = c_1 J_{\frac{1}{4}}(x) + c_2 J_{-\frac{1}{4}}(x)$

☒ $y = c_1 J_{\frac{1}{3}}(x) + c_2 J_{-\frac{1}{3}}(x)$

☐ $y = c_1 J_{\frac{1}{23}}(x) + c_2 J_{-\frac{1}{23}}(x)$

Differential Equations (MTH401)

Question: **12** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

If m and n are non negative integers and $P_n(x)$ is a Legendre's polynomial then

Choices:

☒ $\int_{-1}^1 P_m(x) P_n(x) dx = 0 \text{ for } m \neq n$

☐ $\int_{-1}^1 P_m(x) P_n(x) dx = 0 \text{ for } m = n$

☐ $\int_{-1}^1 P_m(x) P_n(x) dx = 0 \text{ for } m < 0$

☐ $\int_{-1}^1 P_m(x) P_n(x) dx = 0 \text{ for } n < 0$

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Differential Equations (MTH401)

Question: **13** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

If there exists a non-zero solution vector X of the system of linear differential equations such that $AX = \lambda X$. Then the number λ is said to be the

Choices:

Random value

☐

None of them

☐

Matrix value

☐

Eigen value

☐

Differential Equations (MTH401)

Question: **14** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

The differential operator form of the differential equation: $|a|y'' + by' + 2y = \cos x$ is ----.

Choices:

$(|a|D^2 + bD + 2)y = \cos x$

☐

$(aD^2 + bD + 2)y = \cos x$

☐

$(-aD^2 + bD + 2)y = \cos x$

☐

$(bD + 2)y = \cos x$

☐

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Differential Equations (MTH401)

Question: **15** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

$M(x, y)dx + N(x, y)dy$ represents a homogeneous differential equation if
Degree of $M(x, y)$ Degree of $N(x, y)$

Choices:

☐ >

☐

☐ =

☐

☐ <

☐

☐ None of these

☐

Differential Equations (MTH401)

Question: **16** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

$\frac{dy}{dx} + p(x)y = q(x)$
In first order linear differential equation, the functions $p(x)$ and $q(x)$ must be

Choices:

☐ Continuous

☐

☐ Discontinuous

☐

☐ Homogeneous

☐

☐ Non homogeneous

☐

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Differential Equations (MTH401)

Question: 17 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

Differential of $(x+y)$ is ----.

Choices:

☐ $dx+dy$

☐ $dx-dy$

☐ $d(xy)$

☐ $d\left(\frac{x}{y}\right)$

Differential Equations (MTH401)

Question: 18 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

For a 2nd order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 3, W_1 = -\frac{1}{4}$, then $u_1' = \dots$.

Choices:

☐ 12

☐ -12

☐ $\frac{1}{12}$

☐ $-\frac{1}{12}$

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Differential Equations (MTH401)

Question: 19 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

The quantity $Z = \sqrt{X^2 + R^2}$ is called

Choices:

☐ reactance of circuit

☐ impedance of circuit

☐ quasi of circuit

☐ none of them

Differential Equations (MTH401)

Question: 20 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

If the tangent lines of two curves are perpendicular at their point of intersection then both the curves are

Choices:

☐ Orthogonal

☐ Non-Orthogonal

☐ Parallel

☐ None of these

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Differential Equations (MTH401)

Question: 21 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

The linear normal form of $y'' - y' + 6y = 0$, by using $y = x_1$, $y' = x_1' = x_2$ and $y'' = x_2'$, is

Choices:

☐ $x_1' = x_2, x_2' = x_2 - 6x_1$ ←

☐ $x_1 = x_2, x_2' = x_2 + x_1$

☐ $x_2' = x_2$

☐ $x_2' = x_2, x_2' = x_2 - 6$

Differential Equations (MTH401)

Question: 22 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

If $A = [a_{ij}]$ is an $m \times n$ matrix and $B = [b_{ij}]$ is an $n \times p$ matrix, the product AB is an

Choices:

☐ $m \times n$

☐ $n \times p$

☐ $m \times p$ ←

☐ $n \times n$

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Differential Equations (MTH401)

Question: **23** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

Suppose that $y_1, y_2, \dots, y_n$ are n solutions of the homogeneous n th order differential equation.
Then the solution is linearly independent on an interval I if and only if

Choices:

☐ $W(y_1, y_2, \dots, y_n) = 0$

☐ $W(y_1, y_2, \dots, y_n) = \infty$

☐ $W(y_1, y_2, \dots, y_n) \neq 0$

Differential Equations (MTH401)

Question: **24** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

$$A = \begin{bmatrix} 2 & 5 & 6 & 9 \\ -2 & 5 & -1 & 0 \\ 7 & 8 & 1 & 3 \end{bmatrix}$$

Order of matrix is

Choices:

☐ 3×3

☐ 4×4

☐ 3×4

☐ 4×3

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Differential Equations (MTH401)

Question: 25 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

$$X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-2t}, X_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} e^{6t}$$

The vectors are

Choices:

linearly dependent

☐

linearly independent

☐

Differential Equations (MTH401)

Question: 26 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

$\frac{d^2 y}{dx^2} + \omega^2 y = 0$,
subject to $x(0) = x_0$, $x'(0) = 0$, is an example of _____ value problem.

Choices:

initial

☐

boundary

☐

complex

☐

real

☐

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Differential Equations (MTH401)

Question: 27 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

The damping force is directly proportional to the _____.

Choices:

Instantaneous velocity

☐

Instantaneous acceleration

☐

Distance

☐

Time

☐

Differential Equations (MTH401)

Question: 28 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

In Cauchy-Euler Equation, to obtain the auxiliary equation we consider _____.

Choices:

$y = e^{mt}$

☐

$y = e^{2mt}$

☐

$y = e^{\tau}$

☐

$y = e^{m/\tau}$

☐

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Differential Equations (MTH401)

Question: 29 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

$\det(A - \lambda I) = 0$; this equation is called _____.

Choices:

☐ eigen value

☒ characteristic equation

☐ eigen vector

☐ polynomial equation

Differential Equations (MTH401)

Question: 30 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

When the characteristic equation has repeated roots, it may _____ to find n linearly independent eigenvectors of the matrix.

Choices:

☐ possible

☒ not possible

☐ totally possible

☐ always happen

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Differential Equations (MTH401)

Question: 31 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

For finding the general solution of the non-homogeneous system of linear differential equations we need to find _____.

Choices:

☐ complementary solution

☐ particular solution

☐ singular solution

☐ Both complementary and particular solution

Differential Equations (MTH401)

Question: 32 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

Let t_0 denote any point in some interval denoted by I . Then the problem of solving the system of differential equations

$$\frac{dX}{dt} = A(t)X + F(t)$$

Subject to the _____ conditions

$$X(t_0) = X_0$$

Choices:

☐ dirichlet

☐ neumann

☐ initial

☐ boundary

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Differential Equations (MTH401)

Question: 33 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

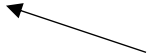
The eigenvalues of homogeneous system

$$X' = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} X$$

are,

Choices:

☐ $\lambda_1 = i, \lambda_2 = -i$

☒ $\lambda_1 = 1, \lambda_2 = -1$ 

☐ $\lambda_1 = 1, \lambda_2 = -i$

☐ $\lambda_1 = -1, \lambda_2 = -i$

Differential Equations (MTH401)

Question: 34 (Marks: 1)

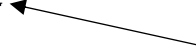
Attempted Questions: 0

Total Questions: 51

If $\begin{vmatrix} 2-\lambda & -8 \\ 2 & -6-\lambda \end{vmatrix} = 0$ then it gives

Choices:

☐ $\lambda = -2$ of multiplicity 2

☒ $\lambda = -6$ of multiplicity 2 

☐ $\lambda = 2$ of multiplicity 2

☐ $\lambda = 6$ of multiplicity 2

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Differential Equations (MTH401)

Question: **35** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

The matrix $\begin{bmatrix} \sqrt{3} & 1 & 2 \\ 0 & \sqrt{3} & 1 \\ 0 & 0 & \sqrt{3} \end{bmatrix}$ has eigenvalue of multiplicity

Choices:

3

☐

2

☐

$\sqrt{3}$

☐

1

☐

Differential Equations (MTH401)

Question: **37** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

The eigen values for the matrix $\begin{pmatrix} 1 & 5 \\ 0 & 7 \end{pmatrix}$ are _____.

Choices:

{1,7}

☐

{1,0}

☐

{1,5}

☐

{6,5}

☐

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Differential Equations (MTH401)

Question: 38 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

The eigen values for the matrix $\begin{pmatrix} 0 & 1 \\ 6 & 7 \end{pmatrix}$ are _____.

Choices:

{0,7}

☐

{1,6}

☐

{7,5}

☐

{-1,-6}

☐

Differential Equations (MTH401)

Question: 39 (Marks: 1)

Attempted Questions: 0

Total Questions: 51

If a matrix $\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$, then eigenvalues are

Choices:

Real and unequal value

☐

Complex eigen value

☐

Repeated & real eigen value

☐

None of them

☐

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Differential Equations (MTH401)

Question: **40** (Marks: 1)

Attempted Questions: **0**

Total Questions: **51**

If a matrix $\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$, then eigenvalues are

Choices:

2,-1

☐

1,-2

☐

2,1

☐

-1,-2

☐

Differential Equations (MTH401)

Question: **41** (Marks: 2)

Attempted Questions: **0**

Total Questions: **51**

What is the form of a power series (x-0)?

Choices:

A standard technique for solving linear differential equations with variable coefficients is to find a solution as an infinite series. Often this solution can be found in the form of a power series.

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Differential Equations (MTH401)

Question: **43** (Marks: 2)

Attempted Questions: **1**

Total Questions: **51**

Define Legendre's polynomial of degree n

Answer:

A second order linear differential equation of the form

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

is called Legendre's differential equation and any of its solution is called Legendre's function. If n is positive integer then the solution of Legendre's differential equation is called a Legendre's polynomial of degree n and is denoted by $P_n(x)$.

Differential Equations (MTH401)

Question: **44** (Marks: 2)

Attempted Questions: **1**

Total Questions: **51**

$$A = \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$$

Find the eigen vector of the matrix corresponding to the eigen value $\lambda = 2$

Answer:

$$\begin{pmatrix} -3 & 1 \\ 2 & -4 \end{pmatrix}$$

A= , corresponding Eigen value $\lambda = -2$.

$$\left(\begin{array}{cc|c} -3-(-2) & 1 & 0 \\ 2 & -4-(-2) & 0 \end{array} \right)$$

$$\left(\begin{array}{cc|c} -1 & 1 & 0 \\ 2 & -2 & 0 \end{array} \right)$$

Add two times row 1 in row 2

$$\left(\begin{array}{cc|c} -1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$$-k_1 + k_2 = 0$$

$$k_1 = k_2$$

Choosing $k_2 = 1$, we get $k_1 = 1$

therefore, eigen vector is $V = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

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Differential Equations (MTH401)

Question: **45** (Marks: 3)

Attempted Questions: **1**

Total Questions: **51**

Solve the system of differential equations

$$\frac{du}{dr} = 3v$$

$$\frac{dv}{dr} = 2u$$

Answer:

$$2 \int y dy = \int x^2 dx$$

$$y^2 = \frac{x^3}{3} + c$$

Differential Equations (MTH401)

Question: **46** (Marks: 3)

Attempted Questions: **1**

Total Questions: **51**

$$X_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-2t} \quad X_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-3t}$$

Find the Wronskian of the solutions and of a system.

Answer:

$$W(X_1, X_2) = \begin{vmatrix} e^{-2t} & 3e^{6t} \\ -e^{-2t} & 5e^{6t} \end{vmatrix} = 8e^{4t} \neq 0, \quad \forall t \in (-\infty, \infty)$$

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Differential Equations (MTH401)

Question: 47 (Marks: 3)

Attempted Questions: 1

Total Questions: 51

What is the significance of r in Frobenius method?

Answer:

When we have a regular singular point $x = x_0$, then we can find at least one series solution of the form $y = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$, where r is the constant that we will determine after solving the differential equation.

Differential Equations (MTH401)

Question: 48 (Marks: 3)

Attempted Questions: 1

Total Questions: 51

Show that $y = c_1 x^{-\frac{1}{2}} + c_2 x^{-\frac{1}{2}} \ln x$ is a solution of $4x^2 \frac{d^2 y}{dx^2} + 8x \frac{dy}{dx} + y = 0$.

Answer:

Solution

The Bessel differential equation is

$$x^2 y'' + xy' + (x^2 - v^2)y = 0 \quad (1)$$

$$x^2 y'' + xy' + \left(x^2 - \frac{1}{4}\right)y = 0 \quad (2)$$

Comparing (1) and (2), we get $v^2 = \frac{1}{4}$, therefore $v = \pm \frac{1}{2}$

So general solution of (1) is $y = C_1 J_{1/2}(x) + C_2 J_{-1/2}(x)$

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Differential Equations (MTH401)

Question: **49** (Marks: 5)

Attempted Questions: **1** Total Questions: **51**

Explain the Ratio Test in Power Series?

Answer:

To determine for which values of x a power series is convergent, one can often use the Ratio Test. The Ratio test states that if

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} c_n (x-a)^n$$

is a power series and

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| |x-a| = L$$

Differential Equations (MTH401)

Question: **50** (Marks: 5)

Attempted Questions: **1** Total Questions: **51**

What is indicial equation and exponent in the differential equation $xy'' + 3y' - y = 0$ with $x=0$ regular singular point?

Answer:

$r(3r - 2) = 0$ [called the indicial equation and its roots $r = \frac{2}{3}, 0$ are called indicial roots or exponents of the singularity.]

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Differential Equations (MTH401)

Question: **51** (Marks: 5)

Attempted Questions: **1** Total Questions: **51**

Show that vectors $X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{-6t}$ and $X_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-2t}$ are linearly dependent.

Answer:

$$W(X_1, X_2) = \begin{vmatrix} e^{-2t} & 3e^{6t} \\ -e^{-2t} & 5e^{6t} \end{vmatrix} = 8e^{4t} \neq 0, \quad \forall t \in (-\infty, \infty)$$

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